Practical 4

Aim - To create a web application to perform **addition of two or three numbers using a SOAP web service**.

1️⃣ SOAP Web Service (Java)

Create new **Web Application** in NetBeans.

Project name: AdditionWS

Create **New → Web Service**

Service Name: AddService

Package: add

AddService.java

```
package add;
import javax.jws.WebService;
import javax.jws.WebMethod;
@WebService(serviceName = "AddService")
public class AddService {
  @WebMethod
  public int addTwo(int a, int b)
  {
    return a + b;
  }
  @WebMethod
  public int addThree(int a, int b, int c)
  {
    return a + b + c;
  }
}
```

Run project.

Check WSDL: http://localhost:8080/AdditionWS/AddService?wsdl

2️⃣ Create Web Service Client

Right click project: New → Web Service Client

WSDL URL: http://localhost:8080/AdditionWS/AddService?wsdl

Finish.

NetBeans generates client classes automatically.

3️⃣ JSP Page (Client)

Create file: add.jsp

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add.jsp code

```
<%@page contentType="text/html"%>
<html>
<body>
<h2>Addition using SOAP Web Service</h2>
<%
try
{
  add.AddService_Service service = new add.AddService_Service();
  add.AddService port = service.getAddServicePort();
  int a = 10;
  int b = 20;
  int c = 30;
  int result1 = port.addTwo(a,b);
  int result2 = port.addThree(a,b,c);
  out.println("Addition of Two Numbers: " + result1);
  out.println("<br>");
  out.println("Addition of Three Numbers: " + result2);
}
catch(Exception e)
{
  out.println(e);
}
%>
</body>
</html>
```

4️⃣ Run JSP

Open browser: http://localhost:8080/AdditionWS/add.jsp

Output example

Addition of Two Numbers: 30

Addition of Three Numbers: 60

Libraries Used

javax.jws.*

JAX-WS

GlassFish Server

No external libraries required.